Adaptive Experimental Design: Posterior CRB, Ignorance Gap, and Adaptive Policies

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Abstract

We develop a self-contained theoretical framework for adaptive experimental design. Starting from the general posterior Cramér–Rao bound (PCRB) and the Tichavský–Muravchik–Nehorai recursion for arbitrary discrete-time state-space models, we successively specialize to a static state, additive Gaussian measurements, and the resulting Bayesian D-optimal problem for *pre-computed* sensor sequences. We then introduce the *ignorance gap* $IG(\boldsymbol{x})$—the log-determinant deficit of the best state-agnostic design relative to an oracle that sees the state—and show that its prior-expectation upper-bounds the improvement achievable by any adaptive policy. When $\mathbb{E}[IG]$ is negligible, the pre-computed design is essentially optimal and no adaptivity is warranted. When $\mathbb{E}[IG]$ is substantial, we define an adaptive policy precisely, contrast it with the pre-computed setting, and characterize the optimal policy via the Bellman equation for the value-to-go W_k. We extend this to the joint optimization of device geometry and adaptive policy using the envelope theorem, prescribe a policy-iteration scheme, and catalog problem classes that exhibit large ignorance gaps. An appendix derives the Bellman equation from first principles.

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1 General PCRB: The Tichavský Recursion

1.1 State-space setting

Consider a discrete-time state-space model:

$$\boldsymbol{x}_k = \boldsymbol{g}_k(\boldsymbol{x}_{k-1}, \boldsymbol{w}_k), \qquad \boldsymbol{w}_k \sim p_{\boldsymbol{w}_k}, \tag{1}$$

$$\boldsymbol{y}_k = \boldsymbol{h}_k(\boldsymbol{x}_k, \boldsymbol{s}_k, \boldsymbol{c}) + \boldsymbol{v}_k, \qquad \boldsymbol{v}_k \sim p_{\boldsymbol{v}_k}, \tag{2}$$

where $\boldsymbol{x}_k \in \mathbb{R}^{d_x}$ is the state, $\boldsymbol{y}_k \in \mathbb{R}^{d_y}$ is the observation, $\boldsymbol{s}_k \in \mathbb{R}^{d_s}$ is a controllable sensor parameter at step k, $\boldsymbol{c} \in \mathbb{R}^{d_c}$ collects device-geometry (structural) parameters held fixed across the horizon, $\boldsymbol{w}_k$ is process noise, and $\boldsymbol{v}_k$ is measurement noise. The prior on the initial state is $\boldsymbol{x}_0 \sim p_0$. The sensor parameter $\boldsymbol{s}_k$ may be fixed at design time or chosen adaptively based on $\boldsymbol{y}_{1:k-1}$.

1.2 The Bayesian Information Matrix

Define the *Bayesian Information Matrix* (BIM) at step k as:

$$\boldsymbol{J}_k \;=\; \mathbb{E}\Big[-\nabla_{\boldsymbol{x}_k}\nabla_{\boldsymbol{x}_k}^{\top} \ln p(\boldsymbol{x}_k \mid \boldsymbol{y}_{1:k})\Big], \tag{3}$$

where the expectation is over the joint distribution $p(\boldsymbol{x}_{0:k}, \boldsymbol{y}_{1:k})$. For any (conditionally or asymptotically) unbiased estimator $\hat{\boldsymbol{x}}_k$ of $\boldsymbol{x}_k$ given $\boldsymbol{y}_{1:k}$, the PCRB is:

$$\mathbb{E}\Big[(\hat{\boldsymbol{x}}_k - \boldsymbol{x}_k)(\hat{\boldsymbol{x}}_k - \boldsymbol{x}_k)^\top\Big] \;\succeq\; \boldsymbol{J}_k^{-1}. \tag{4}$$

1.3 The Tichavský recursion

Theorem 1 (Tichavský, Muravchik & Nehorai, 1998)**.** *Under regularity conditions (twice-differentiable log-densities, finite Fisher information), the BIM satisfies the recursion*

$$\boldsymbol{J}_{k+1} = \boldsymbol{D}_k^{22} - \boldsymbol{D}_k^{21}\left(\boldsymbol{J}_k + \boldsymbol{D}_k^{11}\right)^{-1}\boldsymbol{D}_k^{12} + \boldsymbol{H}_{k+1}, \tag{5}$$

where the $d_x \times d_x$ matrices arise from the state transition and the measurement:

$$\boldsymbol{D}_k^{11} = \mathbb{E}\Big[-\nabla_{\boldsymbol{x}_k}\nabla_{\boldsymbol{x}_k}^\top \ln p(\boldsymbol{x}_{k+1} \mid \boldsymbol{x}_k)\Big], \tag{6}$$

$$\boldsymbol{D}_k^{12} = \mathbb{E}\Big[-\nabla_{\boldsymbol{x}_k}\nabla_{\boldsymbol{x}_{k+1}}^\top \ln p(\boldsymbol{x}_{k+1} \mid \boldsymbol{x}_k)\Big], \tag{7}$$

$$\boldsymbol{D}_k^{21} = \left(\boldsymbol{D}_k^{12}\right)^\top, \tag{8}$$

$$\boldsymbol{D}_k^{22} = \mathbb{E}\Big[-\nabla_{\boldsymbol{x}_{k+1}}\nabla_{\boldsymbol{x}_{k+1}}^\top \ln p(\boldsymbol{x}_{k+1} \mid \boldsymbol{x}_k)\Big], \tag{9}$$

$$\boldsymbol{H}_{k+1} = \mathbb{E}\Big[-\nabla_{\boldsymbol{x}_{k+1}}\nabla_{\boldsymbol{x}_{k+1}}^\top \ln p(\boldsymbol{y}_{k+1} \mid \boldsymbol{x}_{k+1}, \boldsymbol{s}_{k+1})\Big]. \tag{10}$$

The recursion is initialized with the prior information,

$$\boldsymbol{J}_0 = \mathbb{E}\Big[-\nabla_{\boldsymbol{x}_0}\nabla_{\boldsymbol{x}_0}^\top \ln p_0(\boldsymbol{x}_0)\Big] = \boldsymbol{\Sigma}_0^{-1} \qquad \textit{(if } p_0 \textit{ is Gaussian with covariance } \boldsymbol{\Sigma}_0\textit{).} \tag{11}$$

Remark 1. *The $\boldsymbol{D}_k^{ij}$ matrices depend only on the state dynamics; $\boldsymbol{H}_{k+1}$ is the expected measurement Fisher Information Matrix (FIM) of $p(\boldsymbol{y}_{k+1} \mid \boldsymbol{x}_{k+1}, \boldsymbol{s}_{k+1})$ with respect to $\boldsymbol{x}_{k+1}$. The PCRB* (4) *holds for* any *choice of $\boldsymbol{s}_k$—deterministic or adaptively randomized—provided the expectation in $\boldsymbol{H}_{k+1}$ includes the randomness of the sensor policy.*

2 Specialization: Static State

When the state is static ($\boldsymbol{x}_k = \boldsymbol{x}_0$ for all k, i.e., $\boldsymbol{g}_k$ is the identity with vanishing process noise), the transition becomes deterministic: $p(\boldsymbol{x}_{k+1} \mid \boldsymbol{x}_k) = \delta(\boldsymbol{x}_{k+1} - \boldsymbol{x}_k)$. The $\boldsymbol{D}_k^{ij}$ matrices diverge in this limit, and the Schur complement in (5) collapses.

Limiting argument. Consider additive Gaussian dynamics $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \boldsymbol{w}_k$ with $\boldsymbol{w}_k \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q}_k)$ and take $\boldsymbol{Q}_k \to \boldsymbol{0}$. Direct computation from (6)–(9) gives $\boldsymbol{D}_k^{11} = \boldsymbol{D}_k^{22} = \boldsymbol{Q}_k^{-1}$ and $\boldsymbol{D}_k^{12} = \boldsymbol{D}_k^{21} = -\boldsymbol{Q}_k^{-1}$. The recursion becomes

$$\boldsymbol{J}_{k+1} = \boldsymbol{Q}_k^{-1} - \boldsymbol{Q}_k^{-1}\big(\boldsymbol{J}_k + \boldsymbol{Q}_k^{-1}\big)^{-1}\boldsymbol{Q}_k^{-1} + \boldsymbol{H}_{k+1}. \tag{12}$$

Applying the Woodbury identity $\boldsymbol{Q}^{-1} - \boldsymbol{Q}^{-1}(\boldsymbol{J} + \boldsymbol{Q}^{-1})^{-1}\boldsymbol{Q}^{-1} = (\boldsymbol{Q} + \boldsymbol{J}^{-1})^{-1}$ and taking $\boldsymbol{Q}_k \to \boldsymbol{0}$ gives $(\boldsymbol{Q}_k + \boldsymbol{J}_k^{-1})^{-1} \to \boldsymbol{J}_k$. Hence:

$$\boxed{\boldsymbol{J}_{k+1} = \boldsymbol{J}_k + \boldsymbol{H}_{k+1}, \qquad \boldsymbol{J}_0 = \boldsymbol{\Sigma}_0^{-1}.} \tag{13}$$

Additive information. For a static state, information simply accumulates: each measurement contributes its FIM $\boldsymbol{H}_k$ to the running total. After N steps,

$$\boldsymbol{J}_N \;=\; \boldsymbol{\Sigma}_0^{-1} + \sum_{k=1}^{N} \boldsymbol{H}_k(\boldsymbol{s}_k, \boldsymbol{c}), \tag{14}$$

and the PCRB is

$$\mathbb{E}\Big[(\hat{\boldsymbol{x}} - \boldsymbol{x})(\hat{\boldsymbol{x}} - \boldsymbol{x})^\top\Big] \;\succeq\; \left(\boldsymbol{\Sigma}_0^{-1} + \sum_{k=1}^{N} \boldsymbol{H}_k(\boldsymbol{s}_k, \boldsymbol{c})\right)^{-1}. \tag{15}$$

The additive structure holds whether $\boldsymbol{s}_k$ is fixed or chosen adaptively—adaptivity affects the *value* of each $\boldsymbol{H}_k$, not the sum structure.

3 Specialization: Nonlinear Gaussian Measurement

Now suppose the observation model is

$$\boldsymbol{y}_k = \boldsymbol{h}(\boldsymbol{x}, \boldsymbol{s}_k, \boldsymbol{c}) + \boldsymbol{v}_k, \qquad \boldsymbol{v}_k \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{R}_k), \tag{16}$$

where $\boldsymbol{h} : \mathbb{R}^{d_x} \times \mathbb{R}^{d_s} \times \mathbb{R}^{d_c} \to \mathbb{R}^{d_y}$ is a possibly nonlinear measurement function and $\boldsymbol{R}_k$ is the noise covariance (possibly depending on $\boldsymbol{s}_k$ and $\boldsymbol{c}$).

Measurement FIM. The log-likelihood is

$$\ln p(\boldsymbol{y}_k \mid \boldsymbol{x}, \boldsymbol{s}_k) = -\tfrac{1}{2}(\boldsymbol{y}_k - \boldsymbol{h})^\top \boldsymbol{R}_k^{-1}(\boldsymbol{y}_k - \boldsymbol{h}) + \text{const}, \tag{17}$$

and the measurement Jacobian is $\boldsymbol{F}_k = \partial \boldsymbol{h}/\partial \boldsymbol{x}\big|_{\boldsymbol{x},\boldsymbol{s}_k,\boldsymbol{c}} \in \mathbb{R}^{d_y \times d_x}$. A direct calculation (the cross-term in the Hessian vanishes under $\mathbb{E}_{\boldsymbol{y}|\boldsymbol{x}}$) gives the FIM at a specific $\boldsymbol{x}$:

$$\mathrm{FIM}(\boldsymbol{x}; \boldsymbol{s}_k, \boldsymbol{c}) = \boldsymbol{F}_k(\boldsymbol{x})^\top \boldsymbol{R}_k^{-1} \boldsymbol{F}_k(\boldsymbol{x}). \tag{18}$$

Taking the expectation over $\boldsymbol{x}$ (as required by (10)):

$$\boldsymbol{H}_k(\boldsymbol{s}_k, \boldsymbol{c}) = \mathbb{E}_{\boldsymbol{x}}\Big[\boldsymbol{F}_k^\top \boldsymbol{R}_k^{-1} \boldsymbol{F}_k\Big]. \tag{19}$$

For **isotropic** noise $\boldsymbol{R}_k = \sigma^2 \boldsymbol{I}$ this simplifies to $\boldsymbol{H}_k = \sigma^{-2}\,\mathbb{E}_{\boldsymbol{x}}[\boldsymbol{F}_k^\top \boldsymbol{F}_k]$.

Combined static + Gaussian form. Combining Sections 2 and 3, for a static state observed through nonlinear Gaussian measurements with isotropic noise:

$$\boxed{\boldsymbol{J}_N = \boldsymbol{\Sigma}_0^{-1} + \frac{1}{\sigma^2} \sum_{k=1}^{N} \mathbb{E}_{\boldsymbol{x}}\Big[\boldsymbol{F}_k(\boldsymbol{x}, \boldsymbol{s}_k, \boldsymbol{c})^\top \boldsymbol{F}_k(\boldsymbol{x}, \boldsymbol{s}_k, \boldsymbol{c})\Big],} \tag{20}$$

with PCRB

$$\mathbb{E}\big[\|\hat{\boldsymbol{x}} - \boldsymbol{x}\|^2\big] \;\geq\; \mathrm{tr}\big(\boldsymbol{J}_N^{-1}\big). \tag{21}$$

Remark 2 (Two notions of $\boldsymbol{J}_N$)**.** *The BIM* (20) *places* $\mathbb{E}_{\boldsymbol{x}}[\cdot]$ inside *the sum. For design, we will also use the* state-dependent *information*

$$\boldsymbol{J}_N(\boldsymbol{x}, \boldsymbol{s}_{1:N}, \boldsymbol{c}) = \boldsymbol{\Sigma}_0^{-1} + \frac{1}{\sigma^2} \sum_{k=1}^{N} \boldsymbol{F}_k(\boldsymbol{x})^\top \boldsymbol{F}_k(\boldsymbol{x}), \tag{22}$$

which is a random matrix through $\boldsymbol{x}$*. The two are related by* $\mathbb{E}_{\boldsymbol{x}}[\boldsymbol{J}_N(\boldsymbol{x})] = \boldsymbol{J}_N$*, and by Jensen's inequality (concavity of* $\ln\det$*),*

$$\mathbb{E}_{\boldsymbol{x}}[\ln\det \boldsymbol{J}_N(\boldsymbol{x})] \;\leq\; \ln\det \mathbb{E}_{\boldsymbol{x}}[\boldsymbol{J}_N(\boldsymbol{x})]. \tag{23}$$

We use the same symbol $\boldsymbol{J}_N$ *for both and disambiguate by whether an* $\boldsymbol{x}$ *argument is shown.*

4 Pre-Computed Optimal Sensors: Averaged D-Optimality

4.1 The fixed-design problem

Suppose we commit to a sensor sequence $\boldsymbol{s}_{1:N} = (\boldsymbol{s}_1, \ldots, \boldsymbol{s}_N)$ at design time, *before* any observations are collected. Each $\boldsymbol{s}_k$ is a fixed vector in $\mathbb{R}^{d_s}$, *not* a function of anything. Coupled with the device geometry $\boldsymbol{c}$, this defines a *pre-computed* (or *static*, or *non-adaptive*) design.

Since information accumulates additively under (20) and the PCRB depends on $(\boldsymbol{c}, \boldsymbol{s}_{1:N})$ only through $\boldsymbol{J}_N$, a natural design objective is the Bayesian D-optimal criterion:

$$\boxed{\min_{\boldsymbol{c},\, \boldsymbol{s}_1, \ldots, \boldsymbol{s}_N} \mathbb{E}_{\boldsymbol{x}}[-\ln\det \boldsymbol{J}_N(\boldsymbol{x}, \boldsymbol{s}_{1:N}, \boldsymbol{c})]\,.} \tag{24}$$

This is a **single-level** optimization over $d_c + Nd_s$ variables. The objective is the expected log-volume of the per-state Cramér–Rao uncertainty ellipsoid ($\propto (\det \boldsymbol{J}_N)^{-1/2}$).

Remark 3 (Why single-level)**.** *The integrand $\sum_k \boldsymbol{F}_k(\boldsymbol{x}, \boldsymbol{s}_k, \boldsymbol{c})^\top \boldsymbol{F}_k(\boldsymbol{x}, \boldsymbol{s}_k, \boldsymbol{c})$ is a sum of per-step contributions where each $\boldsymbol{s}_k$ appears in only one summand and the summand does not reference any $\boldsymbol{y}_{1:k-1}$—there is no inner optimization or dependence on observations to unroll. The outer minimization over $(\boldsymbol{c}, \boldsymbol{s}_{1:N})$ is therefore flat (single-level) in $d_c + Nd_s$ variables.*

4.2 Gradient structure

Problem (24) is smooth in $(\boldsymbol{c}, \boldsymbol{s}_{1:N})$ when $\boldsymbol{h}$ is smooth. Componentwise, the integrand's gradient at a fixed $\boldsymbol{x}$ is

$$\frac{\partial}{\partial s_{k,i}}(-\ln\det \boldsymbol{J}_N(\boldsymbol{x})) = -\frac{2}{\sigma^2}\,\mathrm{tr}\left(\boldsymbol{J}_N(\boldsymbol{x})^{-1}\,\boldsymbol{F}_k^\top \frac{\partial \boldsymbol{F}_k}{\partial s_{k,i}}\right), \tag{25}$$

using $\partial \ln\det A/\partial A = A^{-\top}$ (which reduces to A^{-1} on the symmetric PD $\boldsymbol{J}_N$). The expected gradient is obtained by taking $\mathbb{E}_{\boldsymbol{x}}[\cdot]$ (Monte Carlo in practice). The gradient with respect to $\boldsymbol{c}$ is analogous, with $\partial \boldsymbol{F}_k/\partial c_j$ propagated through whatever physics model relates the geometry to the measurement Jacobian.

The $\boldsymbol{J}_N^{-1}$ factor couples all measurements: $\partial/\partial s_{k,i}$ of the objective depends on the information from *all other* measurements through $\boldsymbol{J}_N$. This is structural to $\ln\det$ criteria.

4.3 Comparison of alphabetic criteria

Classical optimal-design theory considers several scalar functions of $\boldsymbol{J}_N$:

Criterion	Objective	Interpretation
A-optimal	$\min \mathrm{tr}(\boldsymbol{J}_N^{-1})$	Minimize average variance
D-optimal	$\min -\ln\det \boldsymbol{J}_N$	Minimize log-volume of confidence ellipsoid
E-optimal	$\min \lambda_{\max}(\boldsymbol{J}_N^{-1})$	Minimize worst-case variance

D-optimality has three properties that make it the natural choice:

- **Reparametrization invariance**: $\det \boldsymbol{J}_N$ transforms covariantly under nonsingular changes of coordinates on $\boldsymbol{x}$ (up to the Jacobian determinant), while $\mathrm{tr}(\boldsymbol{J}_N^{-1})$ does not.
- **Balanced information**: it penalizes near-singular $\boldsymbol{J}_N$ much more severely than A-optimality, forcing the sensor sequence to inform *all* state directions rather than piling effort into the most informative one.
- **Information-theoretic meaning**: for Gaussian posteriors, $\frac{1}{2}\ln\det(2\pi e\,\boldsymbol{J}_N^{-1})$ is the differential entropy, so minimizing $-\ln\det \boldsymbol{J}_N$ minimizes the posterior entropy—equivalently, maximizing the mutual information $I(\boldsymbol{x}; \boldsymbol{y}_{1:N})$.

The criteria are related by AM–GM: $d_x^{-1}\,\mathrm{tr}(\boldsymbol{J}_N^{-1}) \geq (\det \boldsymbol{J}_N^{-1})^{1/d_x}$ with equality when $\boldsymbol{J}_N$ is a scalar multiple of $\boldsymbol{I}$. For small d_x they tend to move together.

5 The Ignorance Gap

The pre-computed design (24) hedges against the unknown $\boldsymbol{x}$ by averaging over the prior. How much worse is this hedging than an oracle that sees $\boldsymbol{x}$ and picks sensors specifically for it? The ignorance gap answers this question.

5.1 Oracle and fixed-design values

Define the *oracle* value at a specific state $\boldsymbol{x}$:

$$V_{\text{oracle}}(\boldsymbol{x}) \;=\; \max_{\boldsymbol{s}_{1:N}} \ln\det \boldsymbol{J}_N(\boldsymbol{x}, \boldsymbol{s}_{1:N}, \boldsymbol{c}). \tag{26}$$

This is the best possible log-information if the true $\boldsymbol{x}$ were known at design time. No realizable policy can exceed it, because no realizable policy has access to $\boldsymbol{x}$.

Define the *fixed-design* value:

$$V_{\text{fixed}} \;=\; \max_{\boldsymbol{s}_{1:N}} \mathbb{E}_{\boldsymbol{x}}[\ln\det \boldsymbol{J}_N(\boldsymbol{x}, \boldsymbol{s}_{1:N}, \boldsymbol{c})]\,, \tag{27}$$

with optimal pre-computed sensors $\boldsymbol{s}^{\star}_{1:N}$. This is exactly the (negated) optimal value of (24) for fixed $\boldsymbol{c}$.

5.2 Definition of the ignorance gap

The **ignorance gap at $\boldsymbol{x}$** is

$$\boxed{IG(\boldsymbol{x}) \;=\; V_{\text{oracle}}(\boldsymbol{x}) - \ln\det \boldsymbol{J}_N(\boldsymbol{x}, \boldsymbol{s}^{\star}_{1:N}, \boldsymbol{c}) \;\geq\; 0.} \tag{28}$$

It measures how much better the sensors *could* be if tailored specifically to this $\boldsymbol{x}$, relative to the one-size-fits-all $\boldsymbol{s}^{\star}_{1:N}$. By Jensen's inequality and the definition of V_{fixed}, $\mathbb{E}_{\boldsymbol{x}}[IG(\boldsymbol{x})] = \mathbb{E}[V_{\text{oracle}}] - V_{\text{fixed}}$, the prior-expected ignorance cost.

5.3 The expected ignorance gap bounds adaptation

Introduce (a preview of Section 6) the value of the optimal adaptive policy:

$$V_{\text{adaptive}} \;=\; \max_{\pi}\; \mathbb{E}_{\boldsymbol{x},\boldsymbol{y}}[\ln\det \boldsymbol{J}_N(\boldsymbol{x}, \pi(\boldsymbol{y}_{1:N-1}), \boldsymbol{c})]\,, \tag{29}$$

where the max is over all measurable policies $\pi = \{\pi_k : \boldsymbol{y}_{1:k-1} \mapsto \boldsymbol{s}_k\}$. Since a fixed design is a special (constant) policy, $V_{\text{adaptive}} \geq V_{\text{fixed}}$. Since no policy can beat the oracle, $V_{\text{adaptive}} \leq \mathbb{E}_{\boldsymbol{x}}[V_{\text{oracle}}(\boldsymbol{x})]$. Therefore:

$$\boxed{0 \;\leq\; V_{\text{adaptive}} - V_{\text{fixed}} \;\leq\; \mathbb{E}_{\boldsymbol{x}}[IG(\boldsymbol{x})].} \tag{30}$$

This is the *Expected Value of Perfect Information* (EVPI) bound from decision theory: the expected ignorance gap is an upper bound on the value of the optimal adaptive policy over the best fixed design.

Corollary 1. *If $\mathbb{E}_{\boldsymbol{x}}[IG(\boldsymbol{x})] \approx 0$, then no adaptive policy—no matter how clever—can significantly improve upon the pre-computed design* (24).

5.4 When is the ignorance gap small?

$IG(\boldsymbol{x}) \equiv 0$ exactly when $\operatorname{argmax}_{\boldsymbol{s}_{1:N}} \ln\det \boldsymbol{J}_N(\boldsymbol{x}, \boldsymbol{s}_{1:N}, \boldsymbol{c})$ does not depend on $\boldsymbol{x}$—i.e., the same sensor sequence is optimal regardless of the true state. This holds exactly when $\boldsymbol{F}(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$ is independent of $\boldsymbol{x}$ (linear measurement model).

More generally, $IG(\boldsymbol{x})$ is small when the Jacobian's $\boldsymbol{x}$-dependence is weak relative to its $\boldsymbol{s}$-dependence. A Taylor expansion of a weakly-nonlinear measurement $\boldsymbol{h}(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$ around the prior mean $\boldsymbol{\mu}_0$ gives $\boldsymbol{F}(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c}) = \boldsymbol{F}_0(\boldsymbol{s}, \boldsymbol{c}) + O(\|\boldsymbol{x} - \boldsymbol{\mu}_0\|)$; if the prior standard deviation is small compared to the scale on which $\boldsymbol{F}$ varies with $\boldsymbol{x}$, the oracle-optimal sensors are nearly $\boldsymbol{x}$-independent and $IG \approx 0$.

5.5 Computability of the ignorance gap

The ignorance gap is directly computable from the pre-computed solution with *no adaptive solver required*:

1. Solve (24) to obtain $(\boldsymbol{c}^\star, \boldsymbol{s}^\star_{1:N})$.
2. For a Monte Carlo sample $\{\boldsymbol{x}^{(i)}\}_{i=1}^M$ drawn from the prior:
   (a) Evaluate $\ln\det \boldsymbol{J}_N(\boldsymbol{x}^{(i)}, \boldsymbol{s}^\star_{1:N}, \boldsymbol{c}^\star)$ (the fixed-design log-det at $\boldsymbol{x}^{(i)}$).
   (b) Solve $\max_{\boldsymbol{s}_{1:N}} \ln\det \boldsymbol{J}_N(\boldsymbol{x}^{(i)}, \boldsymbol{s}_{1:N}, \boldsymbol{c}^\star)$ (an Nd_s-variable optimization).
   (c) Compute $IG(\boldsymbol{x}^{(i)})$ as their difference.
3. Report sample statistics: $\widehat{\mathbb{E}}[IG]$, $\widehat{\mathrm{std}}[IG]$, $\widehat{\max}[IG]$, $\Pr[IG > \epsilon]$.

If $\widehat{\mathbb{E}}[IG] \ll 1$ nat, Corollary 1 says we are done: the fixed design is essentially optimal. This offline check costs M small optimizations and should precede any investment in an adaptive solver.

Diagnostic statistics. Beyond the mean, the distribution of $IG(\boldsymbol{x})$ is informative:

Statistic	Interpretation
$\mathbb{E}[IG]$	Upper bound on adaptation value (EVPI, in nats)
$\mathrm{std}[IG]/\mathbb{E}[IG]$	Heterogeneity: large $\Rightarrow$ benefit concentrated on certain $\boldsymbol{x}$
$\max IG$	Worst-case suboptimality of fixed design
$\Pr(IG > \epsilon)$	Fraction of state space where adaptation matters

6 Adaptive Policies

When $\mathbb{E}[IG]$ is *not* negligible, the fixed design leaves value on the table: by observing intermediate $\boldsymbol{y}_k$'s we may be able to infer something about $\boldsymbol{x}$ and then use that information to pick later sensors more informatively. This motivates *adaptive* policies.

6.1 Definition of an adaptive policy

An **adaptive policy** is a sequence of measurable functions

$$\pi = \{\pi_1, \pi_2, \ldots, \pi_N\}, \qquad \pi_k : \mathbb{R}^{d_y(k-1)} \to \mathbb{R}^{d_s}, \tag{31}$$

where $\pi_k(\boldsymbol{y}_{1:k-1})$ is the sensor action chosen at step k as a function of *only* the observations collected *before* step k. By convention π_1 takes no argument (nothing has been observed yet), so $\pi_1 \in \mathbb{R}^{d_s}$ is a fixed vector.

Equivalently, since $\boldsymbol{y}_{1:k-1}$ enters only through the posterior $b_k = p(\boldsymbol{x} \mid \boldsymbol{y}_{1:k-1})$ (the *belief*), we can write $\pi_k(b_k)$—a map from beliefs to sensors.

6.2 Contrast with pre-computed sensors

The difference is operational:

Property	Pre-computed (Section 4)	Adaptive (this section)
$\boldsymbol{s}_k$ is	a fixed vector $\in \mathbb{R}^{d_s}$	a function $\pi_k(\boldsymbol{y}_{1:k-1})$
Chosen	at design time, before any data	at run time, after each observation
Depends on $\boldsymbol{y}_{1:k-1}$?	No	Yes
Search space	$\mathbb{R}^{d_c+Nd_s}$ (finite-dim)	space of functions (infinite-dim)
Information benefit	hedges average-case $\boldsymbol{x}$	can target inferred $\hat{\boldsymbol{x}}$
Upper bound on value	V_{fixed} (27)	$V_{\text{adaptive}} \leq V_{\text{fixed}} + \mathbb{E}[IG]$

The pre-computed setting commits the full sequence $\boldsymbol{s}_{1:N}$ before seeing any $\boldsymbol{y}_k$; the adaptive setting selects $\boldsymbol{s}_k$ after $\boldsymbol{y}_{1:k-1}$ have been revealed. The gap between what each can achieve is upper-bounded by $\mathbb{E}[IG]$.

Remark 4 (Not every run-time computation is "adaptive")**.** *A design that executes sensors $\boldsymbol{s}^\star_{1:N}$ at run time but with* fixed *(pre-computed) values is still a pre-computed design in this sense—even though the sensors are physically deployed during data collection. "Adaptive" refers specifically to whether $\boldsymbol{s}_k$* depends on *$\boldsymbol{y}_{1:k-1}$ as a function, not when the numerical values are computed.*

7 Finding the Optimal Adaptive Policy: Bellman's Equation

Problem (29) is a search over functions, and is at face value intractable. The key simplification is that the posterior b_k is a *sufficient statistic* for the observation history $\boldsymbol{y}_{1:k-1}$—two histories producing the same posterior are equivalent for all future decisions—so we can work with policies on beliefs.

7.1 Value-to-go

Write $\Phi(\boldsymbol{x}, \boldsymbol{s}_{1:N}) = \ln\det \boldsymbol{J}_N(\boldsymbol{x}, \boldsymbol{s}_{1:N}, \boldsymbol{c})$ for the log-information at state $\boldsymbol{x}$ with sensor sequence $\boldsymbol{s}_{1:N}$. Let $b_k = p(\boldsymbol{x} \mid \boldsymbol{y}_{1:k-1})$ denote the belief at step k. Define the **value-to-go** $W_k(b_k)$: the best achievable expected log-information from step k onward, starting at belief b_k and using the optimal continuation policy.

7.2 Bellman recursion

The value-to-go satisfies the **Bellman equation**:

$$W_k(b_k) = \max_{\boldsymbol{s}_k} \mathbb{E}_{\boldsymbol{y}_k|b_k,\boldsymbol{s}_k}[W_{k+1}(b_{k+1})], \qquad k = N, N-1, \ldots, 1, \tag{32}$$

$$W_{N+1}(b_{N+1}) = \mathbb{E}_{\boldsymbol{x}|b_{N+1}}[\Phi(\boldsymbol{x}, \boldsymbol{s}_{1:N})], \tag{33}$$

where b_{k+1} is the Bayesian posterior update of b_k given the newly observed $(\boldsymbol{y}_k, \boldsymbol{s}_k)$. The optimal policy is

$$\pi^\star_k(b_k) \;=\; \operatorname{argmax}_{\boldsymbol{s}_k} \; \mathbb{E}_{\boldsymbol{y}_k|b_k,\boldsymbol{s}_k}[W_{k+1}(b_{k+1})], \tag{34}$$

and the value of the optimal adaptive policy from the prior is

$$V_{\text{adaptive}} \;=\; W_1(p_0). \tag{35}$$

Equations (32)–(33) reduce an infinite-dimensional optimization (over policies = measurable functions) to a sequence of finite-dimensional optimizations (a max over $\boldsymbol{s}_k$ at each step). The price is that W_k is a function on the space of beliefs, which is itself infinite-dimensional; representing W_k exactly is generally impossible except in special cases (linear-Gaussian, finite-state). See Appendix A for the derivation from first principles.

7.3 Oracle value-to-go and the conditional ignorance gap

Analogous to the unconditional ignorance gap, define the *oracle value-to-go* at step k given committed sensors $\boldsymbol{s}_{1:k-1}$:

$$V_k^\star(\boldsymbol{x}) \;=\; \max_{\boldsymbol{s}_{k:N}} \Phi(\boldsymbol{x}, \boldsymbol{s}_{1:k-1}, \boldsymbol{s}_{k:N}). \tag{36}$$

The adaptive policy can never beat the oracle, so $W_k(b_k) \le \mathbb{E}_{\boldsymbol{x}|b_k}[V_k^\star(\boldsymbol{x})]$. The deficit is the **conditional ignorance gap**:

$$\Gamma_k(b_k) \;=\; \mathbb{E}_{\boldsymbol{x}|b_k}[V_k^\star(\boldsymbol{x})] - W_k(b_k) \;\ge\; 0. \tag{37}$$

At $k = 1$ with the prior, $\Gamma_1(p_0) = \mathbb{E}[V_{\text{oracle}}] - V_{\text{adaptive}}$ is the gap that even the *optimal* adaptive policy cannot close. Substituting $W_k = \mathbb{E}[V_k^\star] - \Gamma_k$ into (32) yields a dynamic program on Γ_k—the *gap DP*—which exposes an explicit exploration–exploitation decomposition of the optimal policy. That reformulation is useful for theoretical analysis (bounds on Γ_1, regime classification) but is not required for the remainder of this document; it is standard and available in the optimal-experimental-design literature.

8 Joint Optimization of Geometry and Adaptive Policy

We now combine the two optimizations: over the device geometry $\boldsymbol{c}$ and over the adaptive policy π.

8.1 The joint problem

The full joint problem is

$$\boxed{\max_{\boldsymbol{c}} \; W_1(p_0;\, \boldsymbol{c}),} \tag{38}$$

where $W_1(p_0;\, \boldsymbol{c})$ is the Bellman-optimal value (32) at the prior for geometry $\boldsymbol{c}$, i.e., the joint problem implicitly solves an inner Bellman recursion for each $\boldsymbol{c}$. The geometry enters through every level of the problem: the Jacobians $\boldsymbol{F}_k(\boldsymbol{x}, \boldsymbol{s}, \boldsymbol{c})$, the oracle map $\boldsymbol{s}^\star(\boldsymbol{x};\, \boldsymbol{c})$, the transition of beliefs (via the likelihoods), and the terminal $\Phi(\boldsymbol{x}, \boldsymbol{s}_{1:N}, \boldsymbol{c})$.

8.2 Envelope theorem for the outer gradient

Differentiating (38) naively would require $\partial\pi^\star/\partial\boldsymbol{c}$ (how the optimal policy changes with $\boldsymbol{c}$)—a formidable object. The *envelope theorem* (Danskin) says we don't need it:

$$\boxed{\left.\frac{dW_1}{d\boldsymbol{c}}\right|_{\boldsymbol{c}=\boldsymbol{c}_0} = \left.\frac{\partial}{\partial\boldsymbol{c}}\, \mathbb{E}[\Phi(\boldsymbol{x},\, \pi^\star(\boldsymbol{y};\, \boldsymbol{c}_0),\, \boldsymbol{c})]\right|_{\boldsymbol{c}=\boldsymbol{c}_0},} \tag{39}$$

where $\pi^\star(\cdot;\, \boldsymbol{c}_0)$ is the optimal policy *frozen* at $\boldsymbol{c}_0$, treated as a fixed function of its argument. In words: at the optimal policy for a given geometry, the first-order effect of perturbing the policy vanishes (by the first-order optimality condition on π). We differentiate as if the policy were fixed—only the direct effect of $\boldsymbol{c}$ on the measurement matters, evaluated at whatever sensor values the optimal policy happens to emit.

8.3 Policy iteration scheme

Equation (39) enables an alternating **policy iteration** scheme:

1. **Policy step**: for current $\boldsymbol{c}$, solve or approximate the Bellman recursion (32) to obtain (or approximate) $\pi^\star(\boldsymbol{c})$.

2. **Geometry step**: take a gradient step on $\boldsymbol{c}$ using $\partial\Phi/\partial\boldsymbol{c}$ evaluated at $\pi^\star(\boldsymbol{y};\,\boldsymbol{c})$—the same adjoint/sensitivity computation one would use in the fixed-design problem (24), only now evaluated at the (possibly belief-dependent) sensors produced by the current policy.
3. Repeat until convergence.

Because the Bellman recursion is generically intractable, step 1 is where the computational burden lies; step 2 is cheap by the envelope theorem.

8.4 Reduction by regime

The ignorance gap $\mathbb{E}[IG]$ controls how much of this machinery is actually needed:

Regime	Joint problem reduces to	Solver
$\mathbb{E}[IG] \approx 0$	Fixed D-optimal: $\max_{\boldsymbol{c},\boldsymbol{s}_{1:N}} \mathbb{E}[\Phi]$	Single-level gradient descent
$\mathbb{E}[IG]$ moderate	Certainty-equivalent policy iteration	Oracle map + recursive filter + adjoint
$\mathbb{E}[IG]$ large	Full Bellman DP + policy iteration	Parameterized policy + DP solver

The first column is the problem one *actually* solves; the second is the minimum machinery required. The computability check of Section 5 determines which row applies, *before* any solver is implemented.

Certainty-equivalent (CE) policy. In the moderate-$\mathbb{E}[IG]$ regime, a natural and often near-optimal heuristic is the *certainty-equivalent* policy: at each step, plug the posterior mean $\hat{\boldsymbol{x}}_{k-1} = \mathbb{E}[\boldsymbol{x} \mid b_k]$ into the oracle map and use the resulting sensors,

$$\boldsymbol{s}_k^{\mathrm{CE}} \;=\; \boldsymbol{s}_k^\star(\hat{\boldsymbol{x}}_{k-1}). \tag{40}$$

This replaces the Bellman optimization with a per-step plug-in and is exact as the posterior concentrates; its suboptimality is controlled by the rate of posterior variance reduction and the Lipschitz constant of the oracle map $\boldsymbol{x} \mapsto \boldsymbol{s}^\star(\boldsymbol{x})$.

9 A Tractable Case Study: Adaptive Radar Beam Search

We conclude with a concrete problem that exercises every ingredient of the framework: the ignorance gap is demonstrably large, the geometry $\boldsymbol{c}$ exerts first-order control on achievable performance, *and* the Bellman recursion (32)–(33) is exactly solvable. The policy-iteration scheme of Section 8 therefore becomes a concrete algorithm rather than an aspirational one. The problem is deliberately small and canonical so that every claim can be verified numerically.

9.1 Problem setup

State. A single target occupies one of K discrete cells at distinct angular positions. We use $K = 16$ cells equally spaced on a circular arc at fixed range from the antenna, so that cell $z \in \{1, \ldots, K\}$ has bearing $\varphi(z) = 2\pi(z-1)/K \in [0, 2\pi)$. The prior p_0 on z is uniform (arbitrary but known priors are equally handled).

Actions. At each step $k = 1, \ldots, N$ with N modest (we use $N = 4$), the operator steers the antenna to a pointing direction $\theta_k \in \Theta = \{\theta_1, \ldots, \theta_M\}$ taken from a finite, symmetric set of $M \sim K$ candidate pointings.

Observations. Each pointing returns a single binary detection $y_k \in \{0,1\}$ with Bernoulli likelihood

$$\Pr(y_k = 1 \mid z, \theta_k;\, \boldsymbol{c}) \;=\; p_{\mathrm{FA}} + (p_{\mathrm{D}} - p_{\mathrm{FA}})\, G_{\boldsymbol{c}}(\theta_k - \varphi(z))\,, \tag{41}$$

where $G_{\boldsymbol{c}}(\alpha) \in [0,1]$ is the normalized antenna gain pattern at angular offset α from the steered main-lobe direction, p_{D} is the on-target detection probability, and p_{FA} is the false-alarm rate.

Geometry. The design variable $\boldsymbol{c}$ is the full vector of antenna-array parameters. Concretely, take the antenna to be a phased array of N_e elements at positions $\{\boldsymbol{r}_n\}_{n=1}^{N_e}$ on the aperture plane, driven by complex excitations $\{a_n\}_{n=1}^{N_e} \in \mathbb{C}^{N_e}$ (each $a_n = |a_n| e^{\mathrm{i}\varphi_n}$ carries both an amplitude *and* a phase). The far-field array factor at angular offset α from the steered direction is

$$F_{\boldsymbol{c}}(\alpha) \;=\; \sum_{n=1}^{N_e} a_n \,\exp\!\big(\mathrm{i}\, k_0\, \boldsymbol{r}_n \cdot \hat{\boldsymbol{\alpha}}\big), \qquad k_0 = 2\pi/\lambda, \tag{42}$$

and the normalized gain pattern used in (41) is $G_{\boldsymbol{c}}(\alpha) = |F_{\boldsymbol{c}}(\alpha)|^2 / \max_{\alpha'} |F_{\boldsymbol{c}}(\alpha')|^2$. The design vector is $\boldsymbol{c} = (\{\boldsymbol{r}_n\}, \{a_n\})$, counted over its real degrees of freedom: $d_c \sim 2N_e$ for a linear array with *fixed* (e.g., half-wavelength-spaced) element positions and only complex excitations as free variables, $\sim 3N_e$ if linear-array positions are also free, and $\sim 4N_e$ for a full planar array (element positions in $\mathbb{R}^2$ plus complex amplitudes). A typical $N_e \in \{8, \ldots, 64\}$ gives d_c in the tens to low hundreds. Receiver parameters $(p_{\mathrm{D}}, p_{\mathrm{FA}})$ are held fixed for clarity, but could be appended to $\boldsymbol{c}$ if they are themselves design variables.

Remark 5 (Why complex excitations matter)**.** *Real non-negative amplitudes (pure amplitude tapers) on a symmetric element layout produce conjugate-symmetric patterns $G_{\boldsymbol{c}}(\alpha) = G_{\boldsymbol{c}}(-\alpha)$, so any null at relative offset $+\alpha$ from the main lobe is mirrored at $-\alpha$. Complex a_n break this mirror symmetry: the extra element-wise phase degree of freedom enables shaped asymmetric main lobes and nulls at* arbitrary *relative offsets (not just symmetric pairs)—a qualitatively richer design space than the real-amplitude subspace. Under standard linear-phase steering the entire pattern still rotates rigidly with the pointing θ_k, so these are offsets relative to the main lobe, not angular positions fixed in absolute terms.*

Terminal reward. Take $\Phi(z, \theta_{1:N}) = \ln K - H(z \mid y_{1:N})$, the information gained about z relative to the uniform prior. Equivalently (for a uniform prior) $\Phi = I(z; y_{1:N})$, the mutual information between the unknown cell and the observation sequence, which is also $\ln\det \boldsymbol{J}_N$ up to an additive constant under the categorical's softmax parametrization. Thus $\Phi \in [0, \ln K]$ with $\Phi = 0$ at the prior and $\Phi = \ln K$ at full resolution. Maximizing Φ concentrates the posterior on the true cell.

9.2 The ignorance gap is large

Oracle. Knowing z, the oracle always points $\theta_k^\star = \varphi(z)$ (or the nearest element of Θ), receiving an on-target Bernoulli with probability p_{D} at every step. After N such measurements, the likelihood ratio for cell z against any competing cell grows as $\big(p_{\mathrm{D}}(1-p_{\mathrm{FA}})/(p_{\mathrm{FA}}(1-p_{\mathrm{D}}))\big)^{\#\text{ detections}}$, which concentrates the posterior on z. For $p_{\mathrm{D}} \gg p_{\mathrm{FA}}$ and N at or above the information-theoretic minimum $\log_2 K$ (at the benchmark $K = 16$, $N = 4$ saturates this), the posterior is essentially fully resolved (entropy $\to 0$), giving $V_{\mathrm{oracle}}(z) \approx \ln K$.

Fixed design (narrow-beam family). Assume the geometry $\boldsymbol{c}$ produces a single-main-lobe, directional gain pattern with main-lobe angular width $\Delta\theta_{\mathrm{ML}}(\boldsymbol{c}) \lesssim 2\pi/K$ (one cell per main lobe) and negligible sidelobes. The fixed design must commit $\theta_{1:N}$ before any data; with $N < K$ narrow beams, at most N distinct cells are illuminated. Conditioning on where the target falls:

- target lies in one of the N probed cells (probability N/K): the on-target beam drives $H \to 0$, so $\Phi = \ln K$;
- target lies in the $K - N$ un-probed cells (probability $(K-N)/K$): all N beams return the off-beam Bernoulli and cannot distinguish the un-probed cells, so $H \approx \ln(K-N)$ and $\Phi \approx \ln K - \ln(K-N) = \ln\big(K/(K-N)\big)$.

Averaging:

$$V_{\text{fixed}} \;\approx\; \ln K \;-\; \frac{K-N}{K}\,\ln(K-N). \tag{43}$$

Scaling. Combining the two, for designs with narrow single main lobes and budget $N < K$,

$$\boxed{\mathbb{E}[IG] \;=\; V_{\text{oracle}} - V_{\text{fixed}} \;\approx\; \frac{K-N}{K}\,\ln(K-N)\ \text{nats},} \tag{44}$$

which, for $K \gg N$, scales as $\sim \ln K$; it vanishes continuously as $N \to K$. Note that $\mathbb{E}[IG]$ is always bounded above by the prior entropy $\ln K$, as it must be.

Numerical estimate. Take $K = 16$, $N = 4$, a narrow-beam design with main-lobe width $\Delta\theta_{\text{ML}} \lesssim 2\pi/K = \pi/8$ (e.g., a uniform linear array with $N_e \gtrsim 16$ half-wavelength-spaced elements), $p_{\text{D}} = 0.9$, $p_{\text{FA}} = 0.05$. Equation (44) gives $\mathbb{E}[IG] \approx (12/16)\ln 12 \approx 1.86$ nats, consistent with Monte Carlo evaluation of (28). The EVPI upper bound is substantial—a 2.5× average reduction in the Cramér–Rao uncertainty volume ($e^{\mathbb{E}[IG]/2} \approx e^{0.93}$)—though, as the Rough-sizing discussion below shows, whether adaptation actually *attains* any of that bound is a regime-dependent question, not implied by the size of $\mathbb{E}[IG]$ alone.

9.3 Geometry has first-order impact

The full design $\boldsymbol{c}$ shapes every feature of $G_{\boldsymbol{c}}$ independently: main-lobe width, sidelobe structure, null placement, and grating-lobe behavior. Each of these feeds into the Bernoulli likelihood (41) in a distinct way, and each is independently tunable via the d_c components of $\boldsymbol{c}$:

- *Main-lobe width* $\Delta\theta_{\text{ML}}$ (set primarily by aperture size $L \sim \max_{n,n'} \|\boldsymbol{r}_n - \boldsymbol{r}_{n'}\|$, with $\Delta\theta_{\text{ML}} \sim \lambda/L$ for a well-tapered aperture). Narrow lobes concentrate on-target signal per pointing but can only probe one cell at a time, giving a large $\mathbb{E}[IG]$ in the narrow-beam EVPI sense. Whether adaptation realizes any of that gap is a separate, regime-dependent question analyzed in the Rough-sizing discussion below.
- *First sidelobe level* (controlled by the amplitude taper $\{a_n\}$: uniform taper gives -13 dB, Dolph–Chebyshev or Taylor tapers trade sidelobe level against main-lobe width). Low sidelobes suppress cell-to-cell measurement ambiguity and benefit *both* fixed and adaptive designs.
- *Grating lobes* (set by element spacing: $d > \lambda/2$ introduces copies of the main lobe at other angles). Grating lobes create fundamental ambiguities that no policy can resolve—an aliasing that only $\boldsymbol{c}$ can fix.
- *Null structure at specific relative offsets* (shaped by complex amplitudes and element positions). Deep nulls at designed *relative* offsets from the main lobe allow the measurement to reject specific cell pairs even when the main lobe is steered toward one of them, enhancing cell-pair discrimination in a way that scalar-beamwidth-only parametrizations cannot. Because a phased array steered by linear phase rotates the entire pattern together, these nulls are tied to the main-lobe pointing (their absolute angular positions shift with θ_k); their value is in the pattern's relative structure, not fixed absolute nulls.

Each of these features has first-order impact on both V_{adaptive} and V_{fixed}, and they interact: amplitude tapering that sharpens the main lobe also reduces directivity and raises certain sidelobes. The joint optimum over $(\boldsymbol{c}, \pi)$ is the solution of a d_c-dimensional constrained optimization whose objective is modulated by the adaptive policy that will run on top.

Rough sizing. Binary observations limit the per-step mutual information to at most $\ln 2$ nats, so full resolution of K cells requires at least $N \geq \log_2 K$ measurements regardless of beam design. Under a flat "in-beam / out-of-beam" idealization, write m for the number of cells illuminated per beam; the single-measurement mutual information $I(z; y_k) = H(y_k) - H(y_k \mid z)$ with

$$\Pr(y_k = 1) \;=\; (m/K)\, p_{\mathrm{D}} + ((K-m)/K)\, p_{\mathrm{FA}} \tag{45}$$

is maximized over m at $m^\star$ within $O(1)$ of $K/2$ for any $(p_{\mathrm{D}}, p_{\mathrm{FA}})$ with $p_{\mathrm{D}} > p_{\mathrm{FA}}$. Under this optimum the single-beam main-lobe width is

$$\Delta\theta^\star_{\mathrm{ML}} \;\sim\; \pi \qquad \text{(half the cell ring for } K \text{ cells on a circle).} \tag{46}$$

A key fact about contiguous-arc beams on the ring: each beam has two arc boundaries, so N fixed pointings create at most $2N$ distinct boundary positions and hence at most $2N$ distinguishable segments, each containing $K/(2N)$ cells (in the balanced case). Cells within a segment share the same N-bit label under any *fixed* schedule and cannot be distinguished. Adaptive policies, by contrast, can choose each subsequent pointing so its arc bisects the current posterior support, implementing a binary search that reaches single-cell resolution in $\log_2 K$ steps.

Two regimes follow, at opposite extremes of the beam-width knob:

- *Narrow-beam restriction* ($m \approx 1$, $\Delta\theta_{\mathrm{ML}} \lesssim 2\pi/K$). Each measurement probes one cell. Fixed and adaptive both visit at most N cells in N steps; under the uniform prior they achieve the *same* expected value $V_{\mathrm{fixed}} = V_{\mathrm{adaptive}} \approx \ln K - \frac{K-N}{K}\ln(K-N) \approx 0.91$ nats at $K = 16, N = 4$. The EVPI bound (30) is therefore slack (LHS $= 0$, RHS $= \frac{K-N}{K}\ln(K-N) \approx 1.86$ nats). Large computable $\mathbb{E}[IG]$, zero realized adaptive advantage.
- *Wide-beam regime* ($m \approx K/2$, $\Delta\theta_{\mathrm{ML}} \sim \pi$). Fixed pointings create at most $2N$ segments; at $K = 16, N = 4$ this caps fixed-design resolution at $\ln(2N) = \ln 8 \approx 2.08$ nats. Adaptive binary search reaches full resolution $V_{\mathrm{adaptive}} = \ln K = \ln 16 \approx 2.77$ nats. The adaptive advantage $V_{\mathrm{adaptive}} - V_{\mathrm{fixed}} = \ln(K/(2N)) = \ln 2 \approx 0.69$ nats equals $\mathbb{E}[IG]$ at this $\boldsymbol{c}$, so the EVPI bound is *tight* and adaptation realizes the full gap.

The narrow-beam regime maximizes the *computable* EVPI upper bound but fails to saturate it; the wide-beam regime has a smaller computable $\mathbb{E}[IG]$ but saturates it fully. Joint optimization over $(\boldsymbol{c}, \pi)$ favors the wide-beam design: both $V^\star_{\mathrm{fixed}} = \ln(2N)$ and $V^\star_{\mathrm{adaptive}} = \ln K$ are maximized by $m \approx K/2$. A poor choice of $\boldsymbol{c}$ (sidelobes that confuse specific cell pairs, element spacing introducing grating lobes, or element placement whose $2N$ arc boundaries are badly distributed) wastes the N-step budget. This is exactly the coupling that Section 8's policy iteration is designed to exploit.

9.4 Full Bellman DP is tractable

This is a finite-state, finite-action, finite-observation POMDP with short horizon. Exact solvability follows from three standard facts:

Sondik's theorem (PWLC value functions). For finite POMDPs, each $W_k(b_k)$ is piecewise-linear and convex on the belief simplex. It is represented *exactly* by a finite set of α-vectors $\{\boldsymbol{\alpha}_k^{(i)}\}_{i=1}^{\Gamma_k} \subset \mathbb{R}^K$:

$$W_k(b_k) = \max_i \langle b_k, \boldsymbol{\alpha}_k^{(i)} \rangle. \tag{47}$$

Backward induction from the terminal condition (33)—itself a linear functional of b_{N+1}, hence a single α-vector—produces Γ_{k-1} from Γ_k by a discrete operation (one new vector per $(\theta_k, \boldsymbol{\alpha}_k^{(i_0)}, \boldsymbol{\alpha}_k^{(i_1)})$ triple, modulo dominance pruning).

Dimension count. With $K = 16$, $M = 16$, $|\mathcal{Y}| = 2$, $N = 4$:

- Belief dimension: $K - 1 = 15$.
- Worst-case α-vector growth: $|\Gamma_{k-1}| \leq |\mathcal{A}| \cdot |\Gamma_k|^{|\mathcal{Y}|} = 16 \cdot |\Gamma_k|^2$. Unchecked this is doubly exponential; in practice incremental pruning keeps $|\Gamma_k|$ in the hundreds.
- Memory/time per backup: $O(|\Gamma| \cdot K \cdot M \cdot |\mathcal{Y}|)$—negligible on a laptop.

Off-the-shelf solvers (incremental pruning, SARSOP, PBVI) handle problems of this size in seconds. A hand-coded exact solver in Julia or Python runs in under a minute.

Symmetry reduction. The K-cell ring with a uniform prior and a reflection-symmetric beam pattern carries a dihedral D_K symmetry (cyclic rotations plus one reflection). For $K = 16$ that is $|G| = 32$. The value function inherits the symmetry, $W_k(\sigma \cdot b_k) = W_k(b_k)$ for all $\sigma \in G$, so α-vectors appear in orbits of size up to $|G|$ and only one representative per orbit needs to be explicitly stored. The belief space itself remains $(K-1)$-dimensional—intermediate beliefs generically break the prior's symmetry after the first measurement—but the effective number of distinct α-vectors (and distinct actions, up to symmetry) shrinks by a factor of $|G|$, with corresponding savings in memory and backup time.

In short: step 1 of policy iteration—solve the inner POMDP for the current $\boldsymbol{c}$ (which enters only through the likelihood values $G_{\boldsymbol{c}}(\theta_k - \varphi(z))$ at the $M \cdot K$ action/cell pairs)—is not a research project but a textbook computation. The dependence of the inner DP on $\boldsymbol{c}$ is smooth except at isolated policy-switch boundaries.

9.5 Policy iteration for the joint optimum

With exact inner DP available, the envelope-theorem-based scheme of Section 8 becomes:

1. **Policy step.** For the current design $\boldsymbol{c}_t$, solve the finite POMDP exactly to obtain $\pi^\star(\boldsymbol{c}_t)$ as a depth-N policy tree: every branch corresponds to a value of $y_{1:k-1}$ and maps to an action θ_k.
2. **Geometry step.** For each realized branch of the policy tree, Φ depends on $\boldsymbol{c}$ analytically through the Bernoulli likelihood (41) and the array factor (42) (the derivative of a complex exponential with respect to its amplitude or phase is again a complex exponential). The full vector-valued gradient $\nabla_{\boldsymbol{c}}\Phi \in \mathbb{R}^{d_c}$ is computable in closed form per branch, or by automatic differentiation through the forward model that maps $(\{\boldsymbol{r}_n\}, \{a_n\})$ to the gain values $G_{\boldsymbol{c}}(\theta_k - \varphi(z))$. Aggregate by Monte Carlo over the belief-trajectory distribution induced by $\pi^\star(\boldsymbol{c}_t)$:
$$\nabla_{\boldsymbol{c}} W_1\big|_{\boldsymbol{c}=\boldsymbol{c}_t} \;=\; \mathbb{E}_{z,\, y_{1:N}\sim\pi^\star(\boldsymbol{c}_t)}[\nabla_{\boldsymbol{c}}\Phi(z,\, \pi^\star(y_{1:N};\boldsymbol{c}_t),\, \boldsymbol{c})]_{\boldsymbol{c}=\boldsymbol{c}_t}\,. \tag{48}$$
Because $\pi^\star(\boldsymbol{c}_t)$ is held fixed by the envelope theorem, the right-hand side requires no policy Jacobian $d\pi^\star/d\boldsymbol{c}$—a quantity that is ill-defined at policy-switch boundaries where the argmax in (34) changes.
3. **Gradient step.** Update $\boldsymbol{c}_{t+1} = \mathrm{Proj}_{\mathcal{C}}(\boldsymbol{c}_t + \eta\, \nabla_{\boldsymbol{c}} W_1)$ and repeat, where $\mathcal{C}$ is the feasible set (aperture bounds, per-element power constraints $|a_n| \leq a_{\max}$ or total-power constraint $\sum_n |a_n|^2 \leq P$, minimum-element-spacing to control mutual coupling, etc.) and $\mathrm{Proj}_{\mathcal{C}}$ is a suitable projection. Each complex a_n is parametrized by its real pair $(a_n^{\mathrm{re}}, a_n^{\mathrm{im}}) \in \mathbb{R}^2$, and the gradient is taken with respect to these real components; no Wirtinger calculus is required, and the whole optimization runs in real parameter space $\mathbb{R}^{d_c}$.

Because the inner DP in step 1 is solved *exactly*, there is no approximation error in the gradient direction (48)—only Monte Carlo variance (over z and the stochastic $y_{1:N}$ trajectory), shrinking as $|\text{samples}|^{-1/2}$. Convergence of the outer loop is governed by the smoothness of $W_1(\cdot)$ as a function of $\boldsymbol{c}$, which is continuous except on the codimension-one boundaries where $\pi^\star$ becomes non-unique; the envelope theorem remains valid through these, with a subgradient taking the place of the gradient.

9.6 What this problem illuminates

The radar case study stresses every moving part of the framework at minimum dimension:

- $\mathbb{E}[IG]$ **as a go/no-go gate.** Setting $N = K$ (budget equals number of cells) allows the fixed design to visit every cell, collapsing $\mathbb{E}[IG] \to 0$—the framework correctly predicts that adaptation becomes pointless and the aperture-size optimum degenerates (any aperture large enough to resolve cells suffices). The phase transition at $N = K$ is sharp and numerically verifiable.
- **The DP diagnoses the regime.** The DP produces a different $\pi^\star$ in each of the two beam-width regimes, and the EVPI bound (30) reacts accordingly. In the narrow-beam regime ($m \approx 1$) the DP returns a policy that ties any fixed cell-by-cell schedule, correctly reporting zero adaptive benefit despite a large computable $\mathbb{E}[IG] \approx 1.86$ nats—the EVPI is slack, and the case study shows how to detect a slack EVPI numerically. In the wide-beam regime ($m \approx K/2$) the DP returns a binary-search policy that strictly beats any fixed schedule by the full $\mathbb{E}[IG] = \ln(K/(2N)) \approx 0.69$ nats, saturating the EVPI. Either way the DP is *right about which regime it is in*, and the explore–exploit structure in the wide-beam solution emerges without any hand-crafted policy parametrization, making the case study a discriminating test of the whole framework.
- **Geometry–policy coupling is quantitatively predicted.** The binary-partition main-lobe scaling (46) can be tested numerically against the policy-iteration fixed point; agreement confirms that the envelope-theorem gradient is correctly capturing the joint optimum along the main-lobe-width dimension of $\boldsymbol{c}$. The *remaining* dimensions of $\boldsymbol{c}$—amplitude taper, sidelobe structure, null placement, element positions—are shaped by components of the gradient (48) that no scalar scaling argument can predict, providing a genuine multi-dimensional test of the framework.

Adjacent problems. The same template covers several other canonical testbeds amenable to exact DP:

- *Sequential material identification by spectroscopy*: finite set of candidate chemicals as hypotheses, actions = excitation wavelengths from a discrete grid, observations = quantized absorbance; geometry $\boldsymbol{c}$ = spectrometer linewidth and wavelength grid.
- *Adaptive compressed sensing with known sparsity*: state = support set of $s \ll d$ active indices out of d ambient dimensions (discrete state space of cardinality $\binom{d}{s}$), actions = dictionary rows, observations = quantized linear projections or thresholded detections (finite alphabet); geometry $\boldsymbol{c}$ = dictionary design. For $(d, s) = (10, 2)$ there are 45 states; with observations quantized to a finite alphabet the resulting POMDP remains exactly tractable.
- *Medical-imaging sequential slicing*: discrete lesion-location hypotheses, actions = scan parameters, observations = quantized intensities; geometry $\boldsymbol{c}$ = field-of-view and slice-selection gradients.

All three share the property that finite state/action/observation spaces plus short horizon keep the Bellman recursion exactly solvable, while nonlinear measurement likelihoods keep $\mathbb{E}[IG] > 0$ and the geometry variable meaningful.

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A Derivation of Bellman's Equation

We derive equations (32)–(33) from first principles. The argument is general to any finite-horizon sequential decision problem; we specialize the notation to the setting of this document.

A.1 Problem setup

Recall the ingredients:

- An unknown state $\boldsymbol{x} \sim p_0$ (the prior).
- Horizon N with steps $k = 1, 2, \ldots, N$.
- At step k, an action $\boldsymbol{s}_k \in \mathbb{R}^{d_s}$ is chosen, then $\boldsymbol{y}_k = \boldsymbol{h}(\boldsymbol{x}, \boldsymbol{s}_k, \boldsymbol{c}) + \boldsymbol{v}_k$ is observed, $\boldsymbol{v}_k \sim \mathcal{N}(\boldsymbol{0}, \sigma^2 \boldsymbol{I})$.
- An adaptive policy $\pi = (\pi_1, \ldots, \pi_N)$ with $\pi_k : \boldsymbol{y}_{1:k-1} \mapsto \boldsymbol{s}_k$.
- A terminal reward $\Phi(\boldsymbol{x}, \boldsymbol{s}_{1:N})$ accrued only at the end.

The goal is to find

$$V_{\text{adaptive}} \;=\; \max_{\pi} \; \mathbb{E}_{\boldsymbol{x},\, \boldsymbol{y}_{1:N}}[\Phi(\boldsymbol{x}, \pi(\boldsymbol{y}_{1:N-1}))]\,, \tag{49}$$

a maximization over functions—hopeless at face value.

A.2 Sufficient statistic: the belief state

Because $\boldsymbol{x}$ is the only unknown and the likelihoods $p(\boldsymbol{y}_k \mid \boldsymbol{x}, \boldsymbol{s}_k)$ are known, **all information in $\boldsymbol{y}_{1:k-1}$ relevant to future decisions is carried by the posterior belief**

$$b_k(\boldsymbol{x}) \;=\; p(\boldsymbol{x} \mid \boldsymbol{y}_{1:k-1}). \tag{50}$$

Two observation histories yielding the same b_k are indistinguishable going forward, so we may search over policies $\pi_k : b_k \mapsto \boldsymbol{s}_k$ without loss. Boundary cases: $b_1 = p_0$ (prior, no data yet), b_{N+1} is the posterior after all N observations.

The belief updates by Bayes' rule: given $(b_k, \boldsymbol{s}_k, \boldsymbol{y}_k)$,

$$b_{k+1}(\boldsymbol{x}) \;=\; \frac{p(\boldsymbol{y}_k \mid \boldsymbol{x}, \boldsymbol{s}_k)\, b_k(\boldsymbol{x})}{\int p(\boldsymbol{y}_k \mid \boldsymbol{x}', \boldsymbol{s}_k)\, b_k(\boldsymbol{x}')\, d\boldsymbol{x}'}. \tag{51}$$

The marginal distribution of $\boldsymbol{y}_k$ used for expectations in the Bellman equation is

$$p(\boldsymbol{y}_k \mid b_k, \boldsymbol{s}_k) \;=\; \int p(\boldsymbol{y}_k \mid \boldsymbol{x}, \boldsymbol{s}_k)\, b_k(\boldsymbol{x})\, d\boldsymbol{x}. \tag{52}$$

A.3 Value-to-go: definition

Define the *value-to-go* at step k given belief b_k and committed sensors $\boldsymbol{s}_{1:k-1}$ (the latter suppressed in the notation):

$$W_k(b_k) \;=\; \max_{\pi_k,\pi_{k+1},\ldots,\pi_N} \mathbb{E}[\Phi(\boldsymbol{x},\boldsymbol{s}_{1:N}) \mid b_k,\pi_{k:N}]\,. \tag{53}$$

Boundary cases:

- $k=1$: $W_1(p_0)=V_{\text{adaptive}}$—the quantity we want.
- $k=N+1$: no decisions remain; $\boldsymbol{s}_{1:N}$ are all committed; the terminal value is simply the expected reward under the final belief,

$$W_{N+1}(b_{N+1}) \;=\; \mathbb{E}_{\boldsymbol{x}|b_{N+1}}[\Phi(\boldsymbol{x},\boldsymbol{s}_{1:N})]\,. \tag{54}$$

A.4 Peeling off one step

Start from (53) and split the outer max into "choose $\boldsymbol{s}_k$ now" and "choose the remaining $\pi_{k+1:N}$":

$$W_k(b_k) \;=\; \max_{\boldsymbol{s}_k}\ \max_{\pi_{k+1},\ldots,\pi_N} \mathbb{E}[\Phi \mid b_k,\boldsymbol{s}_k,\pi_{k+1:N}]\,. \tag{55}$$

Apply the *tower property* (law of total expectation) by conditioning on $\boldsymbol{y}_k$:

$$\mathbb{E}[\Phi \mid b_k,\boldsymbol{s}_k,\pi_{k+1:N}] = \mathbb{E}_{\boldsymbol{y}_k|b_k,\boldsymbol{s}_k}[\mathbb{E}[\Phi \mid b_{k+1},\pi_{k+1:N}]]\,, \tag{56}$$

using the fact that b_{k+1} is a deterministic function of $(b_k,\boldsymbol{s}_k,\boldsymbol{y}_k)$ and all remaining randomness depends on b_{k+1} only through the belief (Markov property of beliefs).

Substituting into (55) and pushing $\max_{\pi_{k+1:N}}$ *inside* $\mathbb{E}_{\boldsymbol{y}_k|b_k,\boldsymbol{s}_k}$ (justified below):

$$W_k(b_k) \;=\; \max_{\boldsymbol{s}_k}\ \mathbb{E}_{\boldsymbol{y}_k|b_k,\boldsymbol{s}_k}\left[\max_{\pi_{k+1},\ldots,\pi_N} \mathbb{E}[\Phi \mid b_{k+1},\pi_{k+1:N}]\right]. \tag{57}$$

The inner bracket is exactly $W_{k+1}(b_{k+1})$ by the definition (53) at step $k+1$. Therefore:

$$\boxed{W_k(b_k) \;=\; \max_{\boldsymbol{s}_k}\ \mathbb{E}_{\boldsymbol{y}_k|b_k,\boldsymbol{s}_k}[W_{k+1}(b_{k+1})]\,.} \tag{58}$$

A.5 Why the push-in of the max is legal

In general $\mathbb{E}[\max_i X_i] \geq \max_i \mathbb{E}[X_i]$ (Jensen, since max is convex)—the inequality is strict, not an equality. So why is the move in (58) legal?

Because we are *not* exchanging max with $\mathbb{E}$ abstractly. We are exploiting that $\pi_{k+1:N}$ may *depend on* $\boldsymbol{y}_k$ (through b_{k+1}). A policy that observes $\boldsymbol{y}_k$ before committing the continuation is strictly more powerful than one that commits the continuation in advance—and it is the former that problem (49) allows. Formally, for each realization of $\boldsymbol{y}_k$ we choose the optimal continuation pointwise, and maximizing pointwise inside the expectation gives the same value as maximizing over all measurable continuation plans outside. This is the *interchangeability lemma* for adaptive optimization—the rigorous form of Bellman's *principle of optimality*: an optimal policy's continuation is optimal from whatever state results from its first decision.

A.6 The full Bellman system

Combining the recursion and the boundary condition:

$$W_{N+1}(b_{N+1}) \;=\; \mathbb{E}_{\boldsymbol{x}|b_{N+1}}[\Phi(\boldsymbol{x},\boldsymbol{s}_{1:N})]\,, \tag{59}$$

$$W_k(b_k) \;=\; \max_{\boldsymbol{s}_k}\ \mathbb{E}_{\boldsymbol{y}_k|b_k,\boldsymbol{s}_k}[W_{k+1}(b_{k+1})]\,, \quad k=N,N-1,\ldots,1, \tag{60}$$

$$\pi_k^\star(b_k) \;=\; \operatorname{argmax}_{\boldsymbol{s}_k}\ \mathbb{E}_{\boldsymbol{y}_k|b_k,\boldsymbol{s}_k}[W_{k+1}(b_{k+1})]\,, \tag{61}$$

$$V_{\text{adaptive}} \;=\; W_1(p_0). \tag{62}$$

A.7 Action-value (Q) decomposition

Define the action-value

$$\mathcal{Q}_k(b_k, \boldsymbol{s}_k) \;=\; \mathbb{E}_{\boldsymbol{y}_k|b_k,\boldsymbol{s}_k}[W_{k+1}(b_{k+1})]\,. \tag{63}$$

Then $W_k(b_k) = \max_{\boldsymbol{s}_k} \mathcal{Q}_k(b_k, \boldsymbol{s}_k)$ and $\pi_k^\star = \operatorname{argmax}_{\boldsymbol{s}_k} \mathcal{Q}_k$. This is the standard action-value formulation used in reinforcement learning and approximate dynamic programming.

A.8 Sanity check at the terminal step

At $k = N$, using the terminal (59):

$$W_N(b_N) \;=\; \max_{\boldsymbol{s}_N} \mathbb{E}_{\boldsymbol{y}_N|b_N,\boldsymbol{s}_N}\big[\mathbb{E}_{\boldsymbol{x}|b_{N+1}}[\Phi(\boldsymbol{x}, \boldsymbol{s}_{1:N})]\big] \tag{64}$$

$$\;=\; \max_{\boldsymbol{s}_N} \mathbb{E}_{\boldsymbol{x}|b_N}[\Phi(\boldsymbol{x}, \boldsymbol{s}_{1:N-1}, \boldsymbol{s}_N)]\,, \tag{65}$$

where the second line uses the tower property (for any integrable f not depending on $\boldsymbol{y}_N$, $\mathbb{E}_{\boldsymbol{y}_N|b_N,\boldsymbol{s}_N}\mathbb{E}_{\boldsymbol{x}|b_{N+1}}[f(\boldsymbol{x})] = \mathbb{E}_{\boldsymbol{x}|b_N}[f(\boldsymbol{x})]$). The last step commits no future action, so the optimal $\boldsymbol{s}_N$ is simply the certainty-equivalent choice under the current belief—consistent with the intuition that explore/exploit balance collapses to pure exploit at the final step.

A.9 Why the equation is powerful and cursed

Powerful. The Bellman equation reduces an intractable optimization over *functions* (policies on observation histories) to a recursion over *single-step* decisions. If one can evaluate the max and $\mathbb{E}$ at each step (e.g., by discretizing actions and observations, or by parametric approximation), the problem is solved by N sweeps of dynamic programming.

Cursed. The belief b_k is a probability distribution over $\boldsymbol{x}$—a point in an infinite-dimensional space—and $W_k(\cdot)$ is a function on this space. Representing W_k exactly is generally impossible except in special cases: (i) *linear-Gaussian*, where b_k is fully parametrized by its mean and covariance and W_k admits a closed form via Kalman-filter-based DP; (ii) *finite discrete states*, where b_k is a vector on the simplex and W_k is piecewise-linear and convex (Sondik). For general nonlinear problems, approximations are unavoidable: parametric W_k, particle-filter beliefs, certainty-equivalent policies, or the Jensen-gap bounds of the gap DP.

A.10 Summary

Starting from

$$V_{\text{adaptive}} \;=\; \max_{\pi} \mathbb{E}[\Phi(\boldsymbol{x}, \pi(\boldsymbol{y}_{1:N-1}))]\,, \tag{66}$$

three moves yield the Bellman equation:

(a) **Sufficient statistic reduction**: replace history $\boldsymbol{y}_{1:k-1}$ with belief b_k.
(b) **Peel off one action**: $\max_{\pi_{k:N}} = \max_{\boldsymbol{s}_k} \max_{\pi_{k+1:N}}$.
(c) **Tower property with interchangeability**: push $\max_{\pi_{k+1:N}}$ inside $\mathbb{E}_{\boldsymbol{y}_k}$, revealing $W_{k+1}(b_{k+1})$.

The recursion

$$W_k(b_k) \;=\; \max_{\boldsymbol{s}_k} \mathbb{E}_{\boldsymbol{y}_k|b_k,\boldsymbol{s}_k}[W_{k+1}(b_{k+1})]\,, \qquad W_{N+1}(b_{N+1}) \;=\; \mathbb{E}_{\boldsymbol{x}|b_{N+1}}[\Phi] \tag{67}$$

rests on exactly two facts: **the Markov property of beliefs under Bayesian updating**, and **the principle of optimality** (the optimal continuation can depend on intervening observations without loss).